Gunnar Voet

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Education

2010 PhD in Physical Oceanography, University of Hamburg.
Thesis: *On the Nordic Overturning Circulation*.

2006 Diplom Physik (similar to MSc Physics), University of Hamburg.
Thesis: *Entrainment in the Denmark Strait Overflow Plume by meso-scale Eddies*.

Appointments

2024– Associate Research Scientist at the Scripps Institution of Oceanography, UC San Diego.

2024 Project Scientist at the Scripps Institution of Oceanography, UC San Diego.

2020–2024 Associate Project Scientist at the Scripps Institution of Oceanography, UC San Diego.

2014–2020 Assistant Project Scientist at the Scripps Institution of Oceanography, UC San Diego.

2012–2014 Postdoctoral Researcher at the Applied Physics Laboratory, University of Washington.

2011 Research Associate at the German Federal Maritime and Hydrographic Agency.

2010–2011 Software Developer at the German Federal Waterways Engineering and Research Institute.

2006–2010 Research Associate at the Institute of Oceanography, University of Hamburg.

External Funding

2026–2029 **Voet, G.**, M. H. Alford, A. Le Boyer. Moored Turbulence Observations for TEPEX-E and Beyond. *National Oceanic and Atmospheric Administration*. Lead PI.

2024–2029 Alford, M. H., **G. Voet**, Y-T. Lin, T. Peacock. MURI: Abyssal boundary layers. *Office of Naval Research*. Co-PI.

2024–2027 Alford, M. H., **G. Voet**. Continued analysis of New England Seamounts Acoustics Experiment (NESMA) data. *Office of Naval Research*. Co-PI.

2024–2027 Alford, M. H., **G. Voet**, O. Godin, M. Belal, A. Naveira Garabato. Collaborative research: Direct comparison of ocean temperature and velocity structure from in-situ measurements and distributed optical fiber sensing. *National Science Foundation*. Co-PI.

2023–2025 Couto, N., M. H. Alford, **G. Voet**. Characterizing the warm water flow through Pine Island Thwaites Trough. *National Science Foundation*. Co-PI.

2023–2024 Talley, L., **G. Voet**, A. Waterhouse, C. Whalen. Pilot observations of enhanced near-bottom equatorial turbulence. *National Science Foundation*. Co-PI.

2022–2024 **Voet, G.**, A. Waterhouse. Analysis of temporal and spatial variability of near-inertial energy and flux modified by background vorticity. *Office of Naval Research*. Lead PI.

2021–2026 Whalen, C., A. Waterhouse, **G. Voet**, J. Moum. Collaborative Research: Evaluating mechanisms for enhanced mixing below tropical instability waves. *National Science Foundation*. Co-PI.

2021–2025 Alford, M. H., A. Lucas, **G. Voet**. Shipboard and autonomous measurements of mixed-layer deepening, internal waves and turbulence. *Office of Naval Research*. Co-PI.

2019–2020 **Voet, G.** Moored Oceanographic Instrumentation for Studies of Ocean Dynamics and Turbulence. *Office of Naval Research*. PI.

2018–2023 Ferrari, R., K. L. Polzin, M. H. Alford, **G. Voet**. Collaborative Research: Bottom Boundary Layer Turbulence and the Ocean's Meridional Overturning Circulation. *National Science Foundation*. Co-PI.

2018–2023 Waterhouse, A., **G. Voet**. Near-Inertial Shear and Kinetic Energy in the North Atlantic experiment. *Office of Naval Research*. Co-PI.

2017–2020 **Voet, G.**, M. H. Alford, L. J. Pratt, J. B. Girton, G. S. Carter. Collaborative Research: Hydraulic Processes in the Samoan Passage. *National Science Foundation*. Lead PI.

2015–2019 Alford, M. H., J. A. MacKinnon, **G. Voet**. Flow Encountering Abrupt Topography. *Office of Naval Research*. Co-PI.

Refereed Publications

accepted Cusack, J. M., S. Tan, **G. Voet**, L. J. Pratt, M. H. Alford, G. S. Carter, J. B. Girton. Instabilities of an abyssal overflow within a topographically complex passage. *Journal of Physical Oceanography*.

Wynne-Cattanach, B. L., **G. Voet**, A. C. Naveira Garabato, C. Spingys, H. van Haren, R. Ferrari, M. H. Alford. Boundary-interior exchange quantified with a long-term moored volume budget. *Journal of Physical Oceanography*.

in revision Bellerjeau, C., M. H. Alford, **G. Voet**, N. Couto, A. Le Boyer. Near-inertial wave drag on the Gulf Stream. *Journal of Physical Oceanography*.

Caspar-Cohen, Z., A. F. Waterhouse, **G. Voet**. Impact of mesoscale eddies on near-inertial wave vertical propagation in Iceland Basin. *Journal of Physical Oceanography*.

Voet, G., S. Kawamoto, A. F. Waterhouse, C. B. Whalen. High-resolution moored temperature measurements with commercially available thermistors: Instrumentation, calibration, and drift correction. *Journal of Atmospheric and Oceanic Technology*.

submitted Luecke, C. A., A. E. Simpson, J. D. Nash, J. A. MacKinnon, M. H. Alford, **G. Voet**, B. K. Arbic. Patterns of turbulent dissipation from moored thermistor chains. *Journal of Geophysical Research: Oceans*.

Voet, G., M. H. Alford, C. Bellerjeau. Gulf Stream enhances turbulent mixing and diapycnal upwelling at Atlantis II Seamount. *Journal of Geophysical Research: Oceans*.

2026 Alford, M. H., M. Belal, A. C. Naveira Garabato, C. Spingys, **G. Voet**, O. A. Godin, E. M. Uzhansky, J. Ward Neale, J. Reis, A. Loureiro. Ground-truthed fiber-optic sensing reveals internal tide bores over Madeira's steep slope. *Geophysical Research Letters*, 53(13), e2026GL124117.

Anutaliya, A., C. B. Whalen, **G. Voet**, A.-L. Deppenmeier, A. F. Waterhouse, C. Dias Luko, A. Le Boyer. Observation of Yanai wave excitation by tropical instability waves in the equatorial Pacific. *Geophysical Research Letters*, 53(17), e2026GL124052.

Cimoli, L., A. Mashayek, A. C. Naveira Garabato, C.-c. P. Caulfield, M. H. Alford, L. Baker, B. B. Cael, B. Fernández-Castro, C. de Lavergne, E. Ellison, G. Gebbie, J. Gula, P. Heimbach, J. Lanham, A. Le Boyer, K. Oglethorpe, W. R. Peltier, H. R. Pillar, A. M. Piotrowski, S. G. Purkey, N. Reynard, E. Shuckburgh, M. J. Siegert, A. Silvano, L. C. Skinner, C. P. Spingys, L. D. Talley, J. R. Taylor, **G. Voet**, R. G. Williams, C. Wilson, B. L. Wynne-Cattanach. Climatic reach of small-scale turbulence in the ocean interior. *Nature Communications*, 17(1), 5212.

Ma, Y., R. Ferrari, K. Polzin, M. H. Alford, A. C. Naveira Garabato, **G. Voet**. Standing wave–induced tidal shear in a submarine canyon in the Rockall Trough. *Journal of Physical Oceanography*, 56(3), 665–680.

Moum, J. N., **G. Voet**, A. F. Waterhouse, A. J. Moulin. Wintertime convection in the subpolar North Atlantic. *Journal of Physical Oceanography*, 56(6), 1405–1418.

2025 Alford, M. H., A. Le Boyer, A. S. Ren, **G. Voet**, C. Bellerjeau, C. B. Whalen, B. Hall, N. Couto. Observations of turbulence generated by a near-inertial wave propagating downward in an anticyclonic eddy. *Geophysical Research Letters*, 52(6), e2024GL114070.

Alford, M. H., A. Thode, **G. Voet**, A. B. Laferriere, A. Le Boyer, N. Couto, K. Souhrada, D. Bevans, E. Berkenpas, L. Freeman. A 90-m-tall neutrally buoyant instrument chain for measuring ocean fine structure in a quasi-Lagrangian reference frame. *Journal of Atmospheric and Oceanic Technology*, 42(4), 425–434.

Alford, M. H., B. Wynne-Cattanach, A. Le Boyer, N. Couto, **G. Voet**, C. P. Spingys, B. Fernández-Castro, A. Forryan, A. C. Naveira Garabato, H. van Haren. Buoyancy flux and mixing efficiency from direct, near-bottom turbulence measurements in a submarine canyon. *Journal of Physical Oceanography*, 55(2), 97–118.

Bellerjeau, C., M. H. Alford, A. Le Boyer, G. Dematteis, A. Naveira Garabato, **G. Voet**, N. Couto, B. L. Wynne-Cattanach. Pathways to turbulent dissipation in a submarine canyon. *Geophysical Research Letters*, 52(7), e2024GL113526.

Naveira Garabato, A. C., C. P. Spingys, B. Fernández-Castro, N. Couto, H. F. Drake, A. Forryan, Z. Gao, Y. Ma, H. Mercier, M.-J. Messias, X. Ruan, **G. Voet**, B. L. Wynne-Cattanach, R. Ferrari, M. H. Alford. Connecting mixing to upwelling along the ocean's sloping boundary. *Geophysical Research Letters*, 52(22), e2025GL119186.

Pinner, O., F. Pollmann, M. Janout, **G. Voet**, T. Kanzow. Internal-wave-induced dissipation rates in the Weddell Sea Bottom Water gravity current. *Ocean Science*, 21(2), 701–726.

2024

Burchard, H., M. Alford, M. Chouksey, G. Dematteis, C. Eden, I. Giddy, K. Klingbeil, A. Le Boyer, D. Olbers, J. Pietrzak, F. Pollmann, K. Polzin, F. Roquet, P. S. Saez, S. Swart, L. Umlauf, **G. Voet**, B. Wynne-Cattanach. Linking ocean mixing and overturning circulation. *Bulletin of the American Meteorological Society*, 105(7), E1265–E1274.

Marques, O. B., M. H. Alford, R. Pinkel, J. A. MacKinnon, **G. Voet**, J. M. Klymak, J. D. Nash. Observations of tidally driven turbulence over steep, small-scale topography embedded in the Tasman slope. *Journal of Physical Oceanography*, 54(2), 601–615.

van Haren, H., **G. Voet**, M. H. Alford, B. Fernández-Castro, A. C. Naveira Garabato, B. L. Wynne-Cattanach, H. Mercier, M.-J. Messias. Near-slope turbulence in a Rockall canyon. *Deep Sea Research Part I*, 206, 104277.

Voet, G., A. F. Waterhouse, A. C. Savage, E. Kunze, J. A. MacKinnon, M. H. Alford, J. A. Colosi, H. L. Simmons, T. Klenz, S. M. Kelly, J. N. Moum, C. B. Whalen, R.-C. Lien, J. B. Girton. Near-inertial energy variability in a strong mesoscale eddy field in the Iceland Basin. *Oceanography*.

Wynne-Cattanach, B. L., N. Couto, H. F. Drake, R. Ferrari, A. Le Boyer, H. Mercier, M.-J. Messias, X. Ruan, C. P. Spingys, H. van Haren, **G. Voet**, K. Polzin, A. C. Naveira Garabato, M. H. Alford. Observations of diapycnal upwelling within a sloping submarine canyon. *Nature*, 630(8018), 884–890.

2023

Gutiérrez Brizuela, N., M. H. Alford, S.-P. Xie, J. Sprintall, **G. Voet**, S. J. Warner, K. Hughes, J. N. Moum. Prolonged thermocline warming by near-inertial internal waves in the wakes of tropical cyclones. *Proceedings of the National Academy of Sciences*, 120(26), e2301664120.

Voet, G., M. H. Alford, J. M. Cusack, L. J. Pratt, J. B. Girton, G. S. Carter, J. M. Klymak, S. Tan, A. M. Thurnherr. Energy and momentum of a density-driven overflow in the Samoan Passage. *Journal of Physical Oceanography*, 53(6), 1429–1452.

2022

Boury, S., R. Supekar, E. C. Fine, R. Musgrave, J. B. Mickett, **G. Voet**, P. Odier, T. Peacock, J. A. MacKinnon, M. H. Alford. Observations of double diffusive staircase edges in the Arctic Ocean. *Journal of Geophysical Research*, 127(11), e2022JC018906.

Tan, S., L. J. Pratt, **G. Voet**, J. M. Cusack, K. R. Helfrich, M. H. Alford, J. B. Girton, G. S. Carter. Hydraulic control of flow in a multi-passage system connecting two basins. *Journal of Fluid Mechanics*, 940, A8.

van Haren, H., **G. Voet**, M. H. Alford, D. J. Torres. Internal wave breaking near the foot of a steep East-Pacific continental slope. *Progress in Oceanography*, 205, 102817.

Wynne-Cattanach, B. L., M. H. Alford, J. A. MacKinnon, **G. Voet**. Measurements of turbulence generated by wake eddies near a steep headland. *Journal of Geophysical Research: Oceans*, 127(8), e2022JC018674.

2021 Couchman, M. M. P., B. Wynne-Cattanach, M. H. Alford, C.-c. P. Caulfield, R. R. Kerswell, J. A. MacKinnon, **G. Voet**. Data-driven identification of turbulent oceanic mixing from observational microstructure data. *Geophysical Research Letters*, 48(23), e2021GL094978.

Hamann, M. M., M. H. Alford, A. J. Lucas, A. F. Waterhouse, **G. Voet**. Turbulence driven by reflected internal tides in a supercritical submarine canyon. *Journal of Physical Oceanography*, 51(2), 591–609.

Lele, R., S. G. Purkey, J. D. Nash, J. A. MacKinnon, A. M. Thurnherr, C. B. Whalen, S. Mecking, L. D. Talley, **G. Voet**. Abyssal heat budget in the Southwest Pacific Basin. *Journal of Physical Oceanography*, 51(11), 3317–3333.

Zeiden, K. L., J. A. MacKinnon, M. H. Alford, D. L. Rudnick, **G. Voet**, H. Wijesekera. Broadband submesoscale vorticity generated by flow around an island. *Journal of Physical Oceanography*, 51(4), 1301–1317.

2020 Luecke, C. A., B. K. Arbic, J. G. Richman, J. F. Shriver, M. H. Alford, J. K. Ansong, S. L. Bassette, M. C. Buijsman, D. Menemenlis, R. B. Scott, P. G. Timko, **G. Voet**, A. J. Wallcraft, L. Zamudio. Statistical comparisons of temperature variance and kinetic energy in global ocean models and observations: Results from mesoscale to internal wave frequencies. *Journal of Geophysical Research: Oceans*, 125(5), e2019JC015306.

Voet, G., M. H. Alford, J. A. MacKinnon, J. D. Nash. Topographic form drag on tides and low-frequency flow: Observations of nonlinear lee waves over a tall submarine ridge near Palau. *Journal of Physical Oceanography*, 50(5), 1489–1507.

2019 Andres, M., M. Siegelman, V. Hormann, R. C. Musgrave, S. T. Merrifield, D. L. Rudnick, M. A. Merrifield, M. H. Alford, **G. Voet**, H. W. Wijesekera, J. A. MacKinnon, L. Centurioni, J. D. Nash, E. J. Terrill. Eddies, topography, and the abyssal flow by the Kyushu-Palau Ridge near Velasco Reef. *Oceanography*, 32(4), 46–55.

Carter, G. S., **G. Voet**, M. H. Alford, J. B. Girton, J. B. Mickett, J. M. Klymak, L. J. Pratt, K. A. Pearson-Potts, J. M. Cusack, S. Tan. A spatial geography of abyssal turbulent mixing in the Samoan Passage. *Oceanography*, 32(4), 194–203.

Cusack, J. M., **G. Voet**, M. H. Alford, J. B. Girton, G. S. Carter, L. J. Pratt, K. Pearson-Potts, S. Tan. Persistent turbulence in the Samoan Passage. *Journal of Physical Oceanography*, 49(12), 3179–3197.

Girton, J. B., J. B. Mickett, Z. Zhao, M. H. Alford, **G. Voet**, J. M. Cusack, G. S. Carter, K. A. Pearson-Potts, L. J. Pratt, S. Tan, J. M. Klymak. Flow-topography interactions in the Samoan Passage. *Oceanography*, 32(4), 184–193.

Johnston, S., M. Schönau, T. Paluszkiewicz, J. MacKinnon, B. Arbic, P. Colin, M. Alford, M. Andres, L. Centurioni, H. C. Graber, K. R. Helfrich, V. Hormann, P. F. J. Lermusiaux, R. Musgrave, B. S. Powell, B. Qiu, D. L. Rudnick, H. L. Simmons, L. St. Laurent, E. Terrill, D. S. Trossman, **G. Voet**, H. W. Wijesekera, K. L. Zeiden. FLEAT: A multiscale observational and modeling program to understand how topography affects flows in the Western North Pacific. *Oceanography*, 32(4), 10–21.

MacKinnon, J. A., M. H. Alford, **G. Voet**, K. Zeiden, T. M. S. Johnston, M. Siegelman, S. Merrifield, M. Merrifield. Eddy wake generation from broadband currents near Palau. *Journal of Geophysical Research: Oceans*, 124, 4891–4903.

Pratt, L. J., **G. Voet**, A. Pacini, S. Tan, M. H. Alford, G. S. Carter, J. B. Girton, D. Menemenlis. Pacific abyssal transport and mixing: Through the Samoan Passage versus around the Manihiki Plateau. *Journal of Physical Oceanography*, 49(6), 1577–1592.

Rudnick, D., K. Zeiden, C. Ou, S. Johnston, J. MacKinnon, M. Alford, **G. Voet**. Understanding vorticity caused by flow passing an island. *Oceanography*, 32(4), 66–73.

Siegelman, M., M. A. Merrifield, E. Firing, J. A. MacKinnon, M. H. Alford, **G. Voet**, H. W. Wijesekera, T. Schramek, E. Terrill. Observations of near-inertial surface currents at Palau. *Oceanography*, 32(4), 74–83.

Wagner, G. L., G. Flierl, R. Ferrari, **G. Voet**, G. S. Carter, M. H. Alford, J. B. Girton. Squeeze dispersion and the effective diapycnal diffusivity of oceanic tracers. *Geophysical Research Letters*, 46, 5378–5386.

2018 Thorpe, S. A., J. Malarkey, **G. Voet**, M. H. Alford, J. B. Girton, G. S. Carter. Application of a model of internal hydraulic jumps. *Journal of Fluid Mechanics*, 834, 125–148.

2017 Savage, A. C., B. K. Arbic, J. G. Richman, J. F. Shriver, M. H. Alford, M. C. Buijsman, J. T. Farrar, H. Sharma, **G. Voet**, A. J. Wallcraft, L. Zamudio. Frequency content of sea surface height variability from internal gravity waves to mesoscale eddies. *Journal of Geophysical Research*, 122(3), 2519–2538.

Savage, A. C., B. K. Arbic, M. H. Alford, J. K. Ansong, J. T. Farrar, D. Menemenlis, A. K. O'Rourke, J. G. Richman, J. F. Shriver, **G. Voet**, A. J. Wallcraft, L. Zamudio. Spectral decomposition of internal gravity wave sea surface height in global models. *Journal of Geophysical Research*, 122(10), 7803–7821.

2016 **Voet, G.**, M. H. Alford, J. B. Girton, G. S. Carter, J. B. Mickett, J. M. Klymak. Warming and weakening of the abyssal flow through Samoan Passage. *Journal of Physical Oceanography*, 46(8), 2389–2401.

2015 **Voet, G.**, J. B. Girton, M. H. Alford, G. S. Carter, J. M. Klymak, J. B. Mickett. Pathways, volume transport and mixing of abyssal water in the Samoan Passage. *Journal of Physical Oceanography*, 45(2), 562–588.

2013 Alford, M. H., J. B. Girton, **G. Voet**, G. S. Carter, J. B. Mickett, J. M. Klymak. Turbulent mixing and hydraulic control of abyssal water in the Samoan Passage. *Geophysical Research Letters*, 40(17), 4668–4674.

2010 Fer, I., **G. Voet**, K. S. Seim, B. Rudels, K. Latarius. Intense mixing of the Faroe Bank Channel overflow. *Geophysical Research Letters*, 37, L02604.

Voet, G., D. Quadfasel. Entrainment in the Denmark Strait overflow plume by meso-scale eddies. *Ocean Science*, 6, 301–310.

Voet, G., D. Quadfasel, K. A. Mork, H. Søiland. The mid-depth circulation of the Nordic Seas derived from profiling float observations. *Tellus A*, 62(4), 516–529.

2008 Dickson, R. R., S. Dye, S. Jónsson, A. Köhl, A. Macrander, M. Marnela, J. Meincke, S. Olsen, B. Rudels, H. Valdimarsson, **G. Voet**. The overflow flux west of Iceland: Variability, origins and forcing. In: *Arctic–Subarctic Ocean Fluxes*. Ed. by R. R. Dickson, J. Meincke, P. Rhines. Springer, 443–474.

Software and Data

2026 **Voet, G.**, M. Alford, J. Girton, G. Carter. Samoan Passage abyssal monitoring array 2012–2014. Dryad.

Voet, G., A. Waterhouse, J. Moum, A. Moulin, J. Colosi, H. Simmons. Near-inertial shear and kinetic energy in the North Atlantic experiment (NISKINe): Moored observations. Dryad.

2024 **Voet, G.**, M. Alford, K. Polzin, A. Naveira Garabato, S. Kawamoto, H. Dufel, B. Hogue, B. Wynne-Cattanach, N. Couto, A. Le Boyer, C. Spingys, M.-J. Messias, H. Mercier. Boundary Layer Turbulence and Abyssal Recipes (BLT): Moored Observations. Dryad.

2023 **Voet, G.**, H. Drake, J. Cusack. modscripps/mixsea: v0.1.2. Zenodo.

2022 **Voet, G.**, M. H. Alford, J. B. Girton, G. S. Carter. Samoan Passage Data Archive. Zenodo.

Talks and Posters at International Conferences

2026 Gordon Research Conference on Ocean Mixing.
Turbulent mixing in the central equatorial Pacific (invited talk). South Hadley, Massachusetts, USA.

Ocean Sciences Meeting.
Observational evidence of near-bottom equatorial turbulence (poster). Glasgow, UK.

2024 Gordon Research Conference on Ocean Mixing.
Stirring and mixing of the Gulf Stream at Atlantis II Seamount (poster). South Hadley, Massachusetts, USA.

Ocean Sciences Meeting.
Tidally driven momentum fluxes and their role in turbulent mixing from moored observations in a Rockall Trough canyon (poster). New Orleans, Louisiana, USA.

2023 11th Warnemünde Turbulence Days.
Energy and momentum of a density-driven overflow in the Samoan Passage (talk). Rostock, Germany.

2022 Gordon Research Conference on Ocean Mixing.
Moored observations of tidal mixing and dynamics in a Rockall Trough canyon (poster). South Hadley, Massachusetts, USA.

2020 Ocean Sciences Meeting.
Observations of lee waves generated by topographic form drag on tides and low-frequency flow over a tall submarine ridge near Palau (talk). San Diego, California, USA.

2019 9th Warnemünde Turbulence Days.
Observations of large internal waves and strong turbulence over deep submarine ridges in the Western North Pacific (talk). Vilm, Germany.

27th IUGG General Assembly.
Observations of low frequency flow-generated oceanic lee waves and associated turbulent dissipation over a tall ridge (talk). Montréal, Québec, Canada.

2018 Gordon Research Conference on Ocean Mixing.
Lee wave observations over a tall ridge (poster). Andover, New Hampshire, USA.

2017 49th International Liège Colloquium & 8th Warnemünde Turbulence Days.
Abyssal turbulent mixing in the Samoan Passage (talk). Liège, Belgium.

2016 Ocean Sciences Meeting.
Observed energy and momentum budget of a hydraulically controlled dense overflow in the Samoan Passage (talk). New Orleans, Louisiana, USA.

2015 26th IUGG General Assembly.
Abyssal volume and heat transport through the Samoan Passage: A 16-month timeseries based on recent observations (talk). Prague, Czech Republic.

2014 Ocean Sciences Meeting.
Volume transports and turbulent mixing in the Samoan Passage (poster). Honolulu, Hawaii, USA.

2012 AGU Fall Meeting.
The Samoan Passage Abyssal Mixing Experiment (talk). San Francisco, California, USA.

2010 Euro-Argo User Workshop.
Argo floats in the Nordic Seas (talk). Paris, France.

2009 MOCA-09 Joint Assembly.
The overflow across the Iceland-Faroe Ridge (poster). Montréal, Québec, Canada.

2008 Ocean Sciences Meeting.
The mid-depth circulation of the Nordic Seas from profiling floats (talk). Orlando, Florida, USA.

2007 North Atlantic Subpolar Gyre Workshop.
Stirring and mixing in the Nordic overflow plumes (poster). Kiel, Germany.

2006 Rapid Climate Change International Science Conference.
Stirring and mixing in the Nordic overflow plumes (poster). Birmingham, UK.

Major Research Cruises

29 cruises, 633 days at sea, 4 cruises as Chief Scientist.

	Cruise	Ship	Region	Role or work	Days
2026	DY203	*RRS Discovery*	Madeira	Moorings, acoustics	7
2025	SKQ2025-21S	*R/V Sikuliaq*	Equatorial Pacific at 140°W	Co-PI, mooring lead	24
	DY202	*RRS Discovery*	Madeira	Chief Scientist	6
2024	SKQ2024-17S	*R/V Sikuliaq*	Equatorial Pacific at 140°W	Co-PI, mooring lead	33
	RR2410	*R/V Roger Revelle*	New England Seamount Chain	Co-PI, mooring lead	30
2023	AR73	*R/V Neil Armstrong*	New England Seamount Chain	Co-PI, mooring lead	23
2022	DY153	*RRS Discovery*	Rockall Trough	Co-PI, mooring lead	16
2021	DY138	*RRS Discovery*	Rockall Trough	Co-PI, mooring lead	47
	DY132	*RRS Discovery*	Rockall Trough	Co-PI, mooring lead	40
2020	AR47	*R/V Neil Armstrong*	Iceland Basin	NISKINe lead PI	36
2019	SR1914	*R/V Sally Ride*	Tropical Pacific, Palau	Chief Scientist	14
	AR35-02	*R/V Neil Armstrong*	Iceland Basin	Chief Scientist	10
2018	TN357	*R/V Thomas G. Thompson*	Tropical Pacific, Palau	Mooring lead	30
2017	OC1710A	*R/V Oceanus*	Central Californian Coastal Waters	Moorings	10
	RR17078	*R/V Roger Revelle*	Tropical Pacific, Palau	Chief Scientist	12
2016	RR1607	*R/V Roger Revelle*	Tropical Pacific, Palau	Co-PI, mooring lead	23
2015	SKQ201511S	*R/V Sikuliaq*	Beaufort Sea & Bering Strait	Mooring lead	34
	RR1501, RR1503	*R/V Roger Revelle*	Tasman Sea	CTD, LADCP, moorings	37
2014	PS1415	*R/V Point Sur*	Monterey Bay	Echosounder, CTD	5
	OC1408B	*R/V Oceanus*	Washington Shelf	SWIMS, MMP, moorings	8
	TN305	*R/V Thomas G. Thompson*	Samoan Passage	CTD, LADCP, VMP, moorings	40
2013	MV1306	*R/V Melville*	CLIVAR/Carbon P02E, Honolulu to San Diego	Co-Chief Scientist	24
2012	RR1209	*R/V Roger Revelle*	Samoan Passage	CTD, LADCP, VMP, moorings	42
2008	HM610	*Håkon Mosby*	Faroe-Bank Channel	CTD, LADCP, VMP	10
2007	MSM05-4	*MS Merian*	Irminger Sea / Denmark Strait	CTD, LADCP, moorings	12
2006	D311	*RRS Discovery*	Irminger Sea / Denmark Strait	CTD, vmADCP, moorings	16
2005	A0805	*Árni Friðriksson*	Irminger Sea / Denmark Strait	CTD, moorings	15
2002	44/02/05	*FS Alexander von Humboldt*	North Atlantic	CTD	29

Mentoring

Students and postdocs

2025– Deena Lang, graduate student in the SIO Physical Oceanography program (advisor).

2025– Zachary Taebel, postdoctoral researcher.

2024–2026 Karl-Erik Paulsson, MS student in Physical Oceanography (advisor).

2017–2020 Jesse Cusack, postdoctoral researcher. Now Assistant Professor at Oregon State University.

Thesis committees

2025– Caique Dias Luko, PhD committee member, SIO Physical Oceanography program.

2026– Caeli Griffin, PhD committee member, SIO Physical Oceanography program.

2026 Zachary Kreitzer, MS committee member, Applied Ocean Science.

2022–2026 Ole Pinner, PhD committee member, Alfred Wegener Institute and University of Bremen.

Professional Services

2025– Associate Editor for *Journal of Physical Oceanography*.

Reviewer for *Journal of Physical Oceanography, Journal of Geophysical Research, Geophysical Research Letters, Ocean Modelling, Ocean Science, Deep Sea Research, Journal of Marine Research, Journal of Marine Systems, Oceanography, Nature Communications*.

Reviewer of research proposals for *NSF* and *NOAA*.

Stipends and Awards

2007 *Annette-Barthelt-Preis für Meeresforschung* for an outstanding thesis in the field of seagoing ocean research, including a research grant from the German Federal Ministry of Education and Research (BMBF).

Other Qualifications

2023 Optimizing Research Mentoring Relationships – Professional Development Workshop.

2021 UC Fieldwork Toolkit Leadership Training Series.

2014 UNOLS Chief Scientist Training Workshop.